

Mahdi Babaei

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EDUCATION

Stevens Institute of Technology | Hoboken, NJ Sep 2023 - Expected May 2027
Ph.D. in Biomedical Engineering | **Focus:** AI-driven healthcare systems, medical image/video processing

University of Qom | Qom, Iran Sep 2018 - May 2021
M.Sc. in Information Technology Engineering | **Focus:** Deep learning for polypharmacy interaction analysis

University of Semnan | Semnan, Iran Sep 2014 - Sep 2018
B.Sc. in Information Technology Engineering | **Focus:** Android and iOS application development

WORK EXPERIENCE

Stevens Institute of Technology | Hoboken, NJ Sep 2023 - Present
Research Assistant

- AI-driven healthcare applications.
- Object tracking and segmentation.
- Transformers, self-supervised learning techniques, and large language models.

Ariasamaneh. | Tehran, Iran May 2021 - Sep 2022
Machine Learning Engineer

- Developed object segmentation and tracking pipelines for real-time video processing.
- Optimized GPU-based pipelines for high performance.
- Implemented multi-threaded data pipelines to handle high throughput video streams (latency -25%).

Ariasamaneh. | Tehran, Iran March 2019 - April 2020
Software Developer & Technical Support

- Android and iOS software development (messenger).
- Cloud manager assistant.

RECENT FIRST-AUTHOR RESEARCH PAPERS

Title: Robust classification of noisy labels using a new hybrid loss function
(Under Review - *Journal of Critical Reviews in Food Science and Nutrition*, 2025)

Title: Self-supervised refinement of cell segmentation leveraging spatiotemporal consistency
(Under Review - *International conference on medical image computing*, 2025)

Title: Automated cell viability measurements from 3D bioprinted tissue samples via deep learning and OCT
(Accepted - *Biomedical Optics Express*, 2025)

SKILLS

Programming: Python, C/C++, SQL, Docker

Machine Learning: Deep Learning (CNNs, RNNs), Generative Models, Large Language Models, Self-Supervised Learning, Computer Vision, Traditional Algorithms

Frameworks / Tools: TensorFlow, PyTorch, Torchvision, Scikit-learn, OpenCV, Hugging Face Transformers

Data Analysis / Visualization: Pandas, Numpy, Matplotlib, Seaborn, Plotly

SELECTED PROJECTS

- * Self-Supervised Segmentation and Tracking (LLM-Assisted)
- * Generative Models for Tissue Simulation
- * Proactive Dialogue Generator for Efficient Clinical Information Collection Using Large Language Model
- * Quantum Computing and Quantum ML
- * Few-Shot Learning for Drug and Agriculture Research